Statistical Analysis of High Blood Pressure and Related Factors in Bangladesh: Evidence from BDHS 2022

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1. Objective

The objective of this study is to analyze data from Bangladesh Demographic and Health Survey (BDHS) 2022 using SPSS in order to describe the distribution of prime variables, summarize age and systolic blood pressure, examine their variability and shape, study the relationship between high blood pressure and selected socio-demographic factors and determine whether age has an effect on systolic pressure.

2. Introduction

High blood pressure is a vital public health problem in a developing country like Bangladesh and is a major risk factor for diseases like heart diseases and stroke. With the increasing prevalence of this sort of non-communicable diseases, it becomes essential to understand the patterns and factors related to blood pressure in the population.

The Bangladesh Demographic and Health Survey (BDHS) 2022 is a nationally representative survey which provides well grounded data on health, demographic and socio-economic characteristics of the Bangladeshi population. This offers an important opportunity to examine patterns of systolic blood pressure and high blood pressure across different socio-demographic, behavioral and health related groups. Hence, statistical analysis of such data helps identify high-risk populations and uniform public health interventions.

In this study, SPSS analyzes the selected variables from the BDHS 2022 dataset. This analysis includes frequency distributions, graphical representation, computation of summary statistics for age and systolic blood pressure, assessment of variability, skewness shape and evaluation of associations between high blood pressure and selected variables using contingency tables and Chi-square tests. Moreover, correlation and regression analysis is also conducted to explore the clinical relevance between age and systolic blood pressure. Through these analyses, the study aims to provide a comprehensive statistical rundown of blood pressure patterns and their associated factors in Bangladesh.

3. Specification table

Field of Study	**Analysis**
Types of data	Quantitative (continuous numerical value) and Qualitative (categorical value)
Data Collection	Secondary date obtained from the Bangladesh Demographic and Health Survey (BDHS) 2022.
Data Format	SPSS data file.
Parameters for Data Collection	Demographic variables (age, marital status, education, division, residence, wealth index), health indicators (systolic blood pressure, high blood pressure status, diabetes status), lifestyle factors (smoking status, consumption of caffeinated drinks)
Description of Data Collection	Data were collected through nationally representative household surveys using standardized questionnaires and physical measurements conducted by trained personnel across Bangladesh. Besides, blood pressure is measured following standard clinical protocols.
Used software	IBM SPSS Statistics.
Data Identification No.	N/A
Supporting Data Information	Yes (survey questionnaires, data dictionary and metadata provided by the DHS program).
Direct URL to Data	https://dhsprogram.com/

4. Methodology

4.1. Data Collection

The study used secondary data obtained from the Bangladesh Demographic and Health Survey (BDHS) 2022. The BDHS collects information on demographic characteristics and health factors using ordered surveys and standardized physical measurements conducted by trained personnel. Consequently, blood pressure measurements were obtained following established clinical protocols to ensure accuracy and reliability.

4.2. Data Preparation

The raw dataset from the Bangladesh Demographic and Health Survey (BDHS) 2022 was imported into IBM SPSS Statistics. Data cleaning was performed to ensure that all 10,086 cases used for analysis were valid. For quantitative variables like Systolic Blood Pressure, extreme values were noted for further distributional assessment.

4.3. Variable Selection

The data was modified in SPSS before analysis. The categorical variables (like division, type of place of residence, wealth index combined, highest educational level attained, had caffeinated drink, smoking status , current marital status, ever told by a doctor/nurse to have diabetes and told by a doctor to have high blood pressure) were converted into numeric codes for analysis. Two primary quantitative variables were selected for central tendency and dispersion analysis:

- Age of household members: Measured in years.
- Systolic blood pressure (SBP): Measured in mmHg.

4.4. Analytical Tools Used

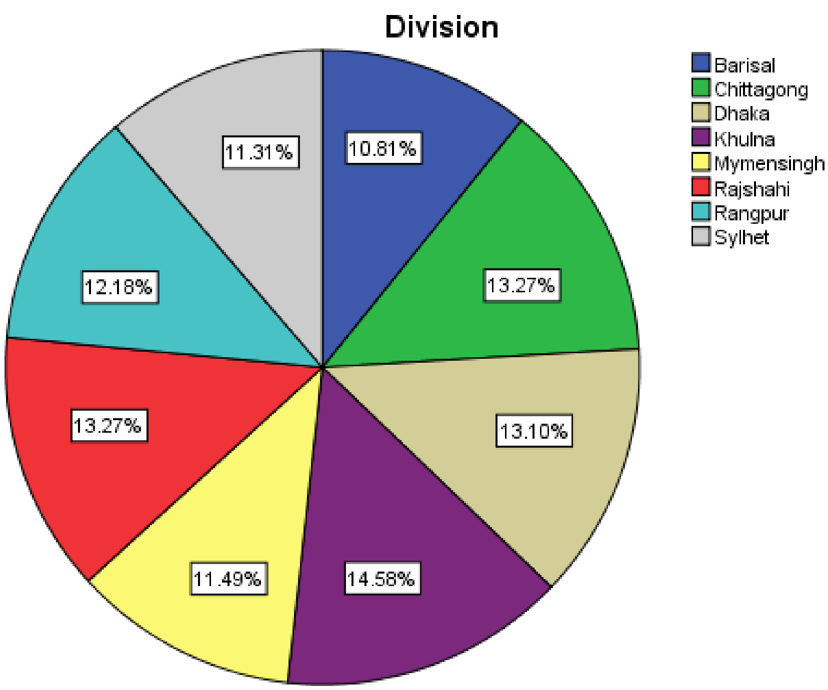
The descriptive statistics were computed to understand the distribution of variables, age of household members and systolic blood pressure measurement. Frequency distribution and graphical representation of each variable were reported. The mean, five number summaries and standard deviation were employed for continuous variables such as age and systolic blood pressure. Several visualisations were created using SPSS to help with numerical descriptions. To visually represent the distributions, pie charts were created for categorical variables like division, type of place of residence, wealth index combined, highest educational level attained, had caffeinated drink, smoking status, current marital status, ever told by a doctor or nurse to have diabetes and high blood pressure to visualize the spread and patterns within the data. Chi-square tests and cross-tabulations (crosstabs) were used to investigate the connection between high blood pressure and other categorical variables, including division, residence, current marital status, diabetes status, wealth index combined, educational status, had caffeinated drinks, and smoking status. Statistical relationships were determined at a significance level of $p < 0.05$. Additionally, the frequency of the variables age and systolic blood pressure were generated using bar charts.

5. Descriptive Statistics

5.1. Frequency Tables:

Division

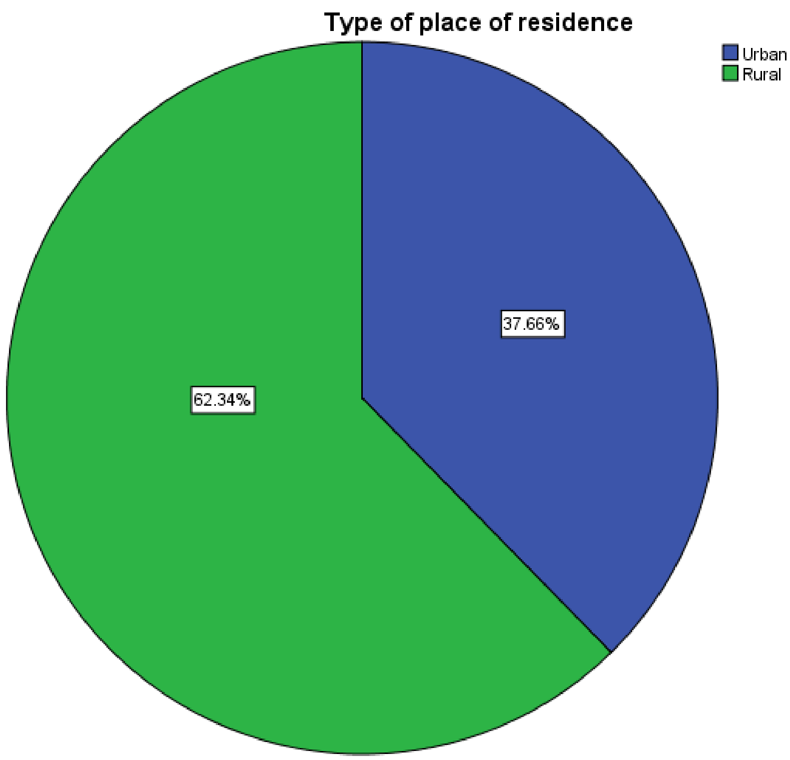
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Barisal	1090	10.8	10.8	10.8
	Chittagong	1338	13.3	13.3	24.1
	Dhaka	1321	13.1	13.1	37.2
	Khulna	1471	14.6	14.6	51.8
	Mymensingh	1159	11.5	11.5	63.2
	Rajshahi	1338	13.3	13.3	76.5
	Rangpur	1228	12.2	12.2	88.7
	Sylhet	1141	11.3	11.3	100.0
	Total	10086	100.0	100.0	



The study analyzed a total of 10,086 participants across eight divisions,1,471 participants were from Khulna, 1,338 from Chittagong, 1,338 from Rajshahi, 1,321 from Dhaka, 1,228 from Rangpur, 1,159 from Mymensingh, 1,141 from Sylhet, and 1,090 from Barisal. In terms of percentage, it can be observed that the majority of participants were from Khulna at 14.6%, followed by Chittagong at 13.3%, Rajshahi at 13.3%, Dhaka at 13.1%, Rangpur at 12.2%, Mymensingh at 11.5%, Sylhet at 11.3%, and the lowest category by Barisal at 10.8%.

Type of place of residence

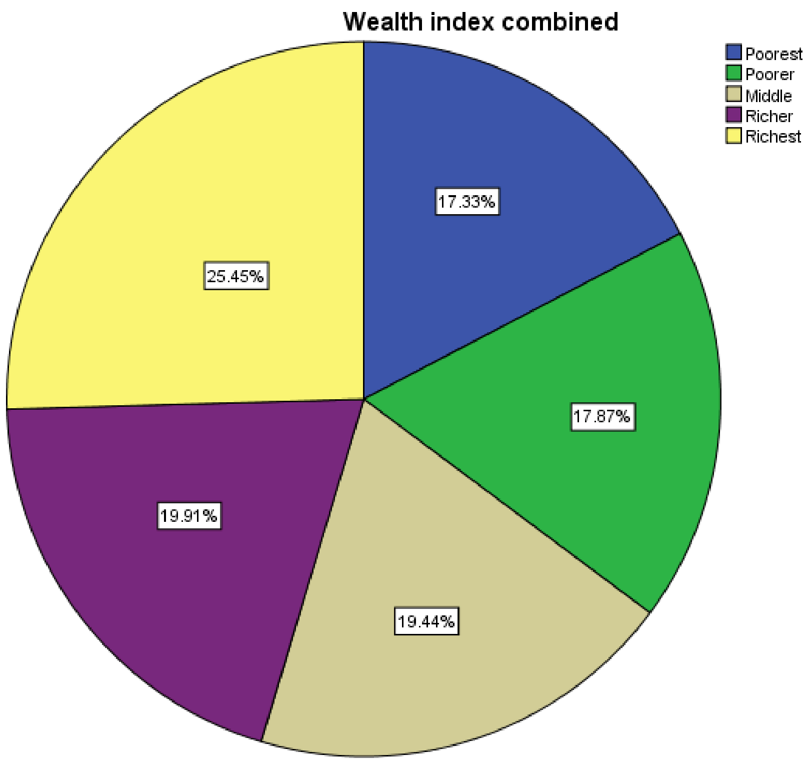
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Urban	3798	37.7	37.7	37.7
	Rural	6288	62.3	62.3	100.0
	Total	10086	100.0	100.0	



In terms of place of residence, out of 10,086 participants, 6,288 lived in rural areas and 3,798 lived in urban areas. In terms of percentage, it can be observed that the majority of participants resided in rural areas at 62.3%, while the urban category accounted for 37.7%.

Wealth index combined

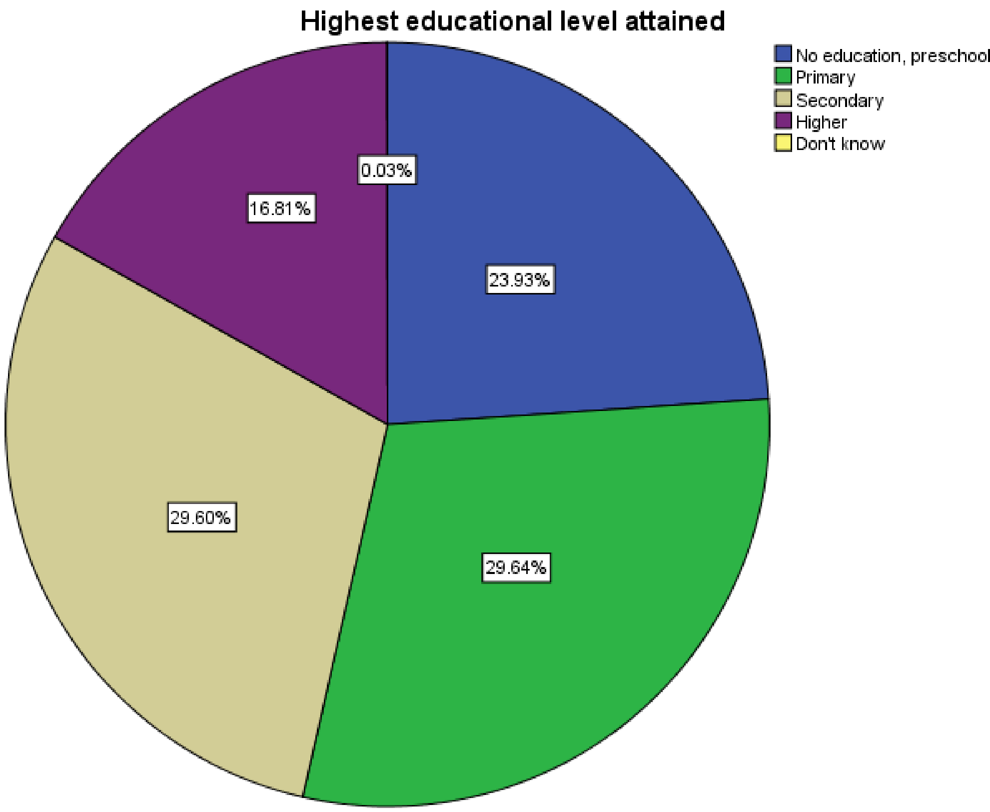
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Poorest	1748	17.3	17.3	17.3
	Poorer	1802	17.9	17.9	35.2
	Middle	1961	19.4	19.4	54.6
	Richer	2008	19.9	19.9	74.5
	Richest	2567	25.5	25.5	100.0
	Total	10086	100.0	100.0	



In terms of wealth index, out of 10,086 participants, 1,748 were poorest, 1,802 were poorer, 1,961 were middle, 2,008 were richer, and 2,567 were richest. In terms of percentage, it can be observed that the largest group was the richest category 25.5%, followed by richer at 19.9%, middle at 19.4%, poorer at 17.9%, and the lowest category was poorest at 17.3%.

Highest educational level attained

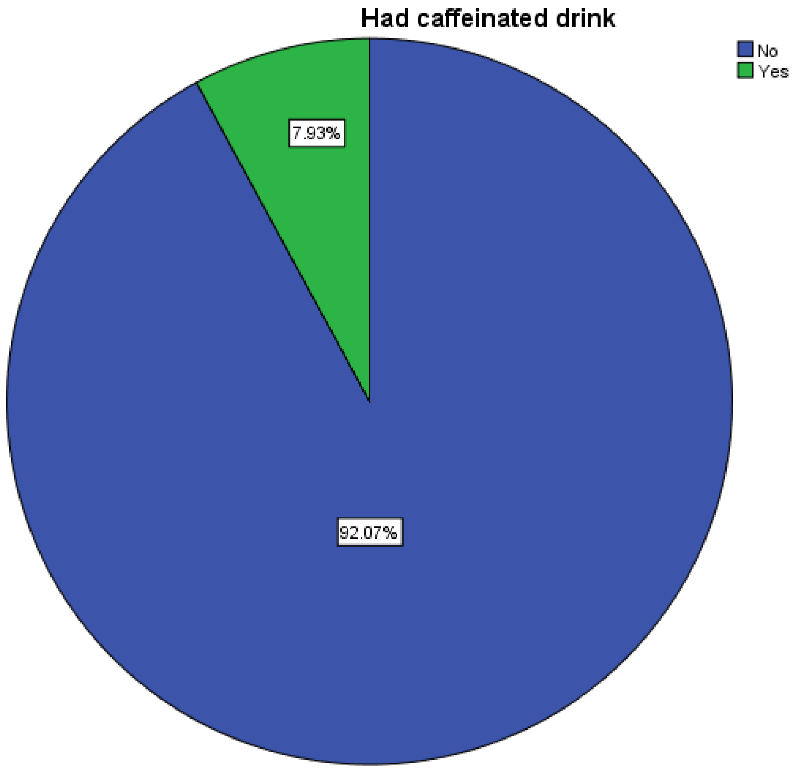
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No education, preschool	2414	23.9	23.9	23.9
	Primary	2989	29.6	29.6	53.6
	Secondary	2985	29.6	29.6	83.2
	Higher	1695	16.8	16.8	100.0
	Don't know	3	.0	.0	100.0
	Total	10086	100.0	100.0	



In terms of highest educational level attained, out of 10,086 participants, 2,414 had no education or preschool, 2,989 had primary education, 2,985 had secondary education, 1,695 had higher education, and 3 did not know. In terms of percentage, it can be observed that the majority of participants were nearly tied between primary education at 29.6% and secondary education at 29.6%, followed by no education at 23.9%, higher education at 16.8%, and the lowest category by "Don't know" at 0.0%.

Had caffeinated drink

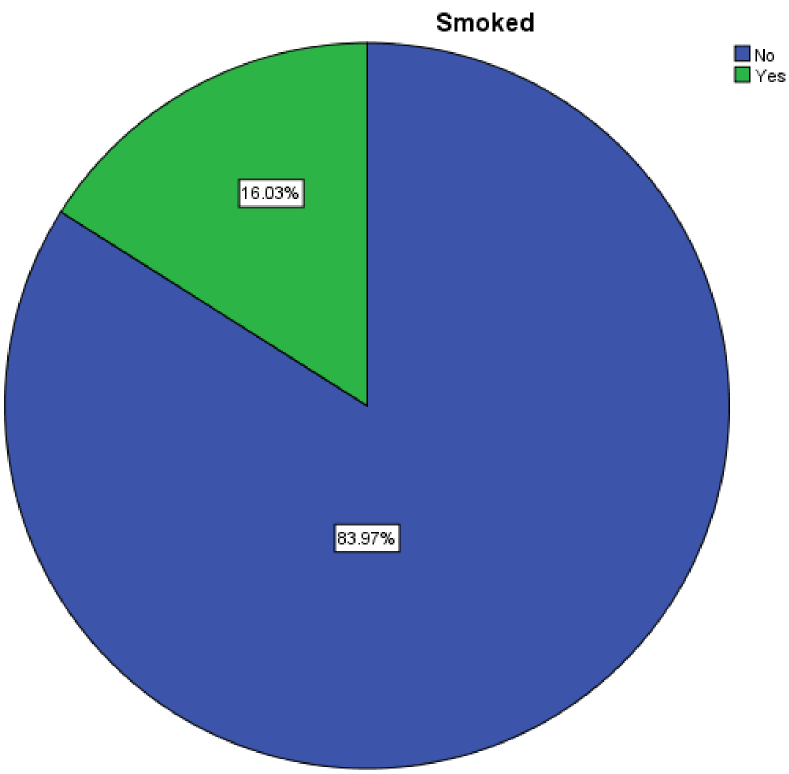
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	9282	92.0	92.1	92.1
	Yes	799	7.9	7.9	100.0
	Total	10081	100.0	100.0	
Missing	9	2	.0		
	System	3	.0		
	Total	5	.0		
Total		10086	100.0		



In terms of caffeinated drink consumption, out of 10,086 participants, 9,282 had not consumed a caffeinated drink and 799 had. In terms of percentage, it can be observed that the vast majority of participants had not consumed a caffeinated drink at 92.0%, while the "Yes" category accounted for 7.9%.

Smoked

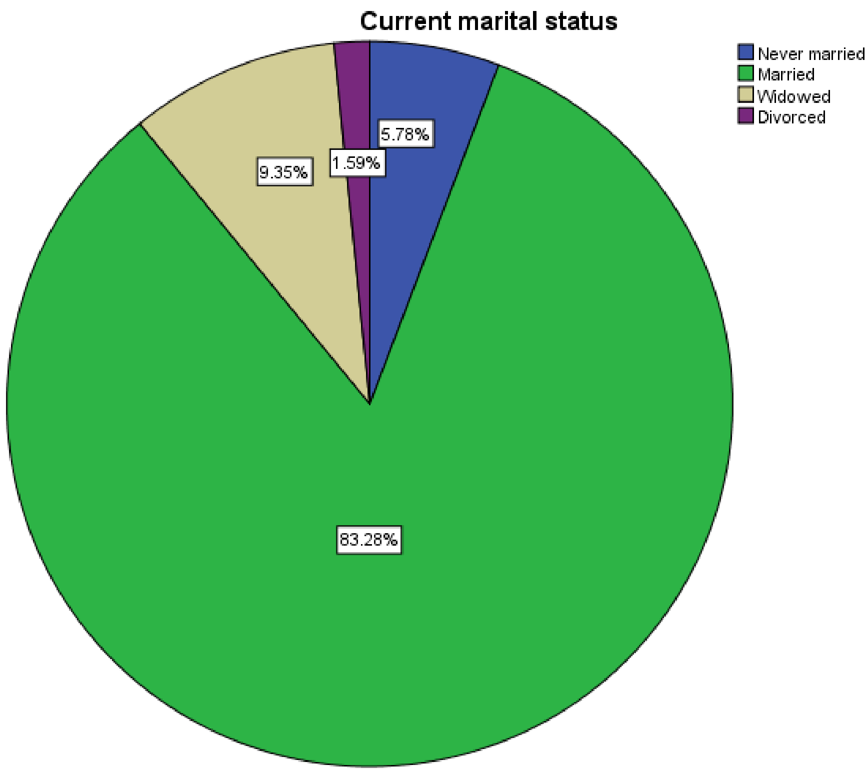
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	8464	83.9	84.0	84.0
	Yes	1616	16.0	16.0	100.0
	Total	10080	99.9	100.0	
Missing	9	3	.0		
	System	3	.0		
	Total	6	.1		
Total		10086	100.0		



In terms of smoking habits, out of 10,086 participants, 8,464 had not smoked and 1,616 had. In terms of percentage, it can be observed that the vast majority of participants had not smoked at 83.9%, while the "Yes" category accounted for 16.0%.

Current marital status

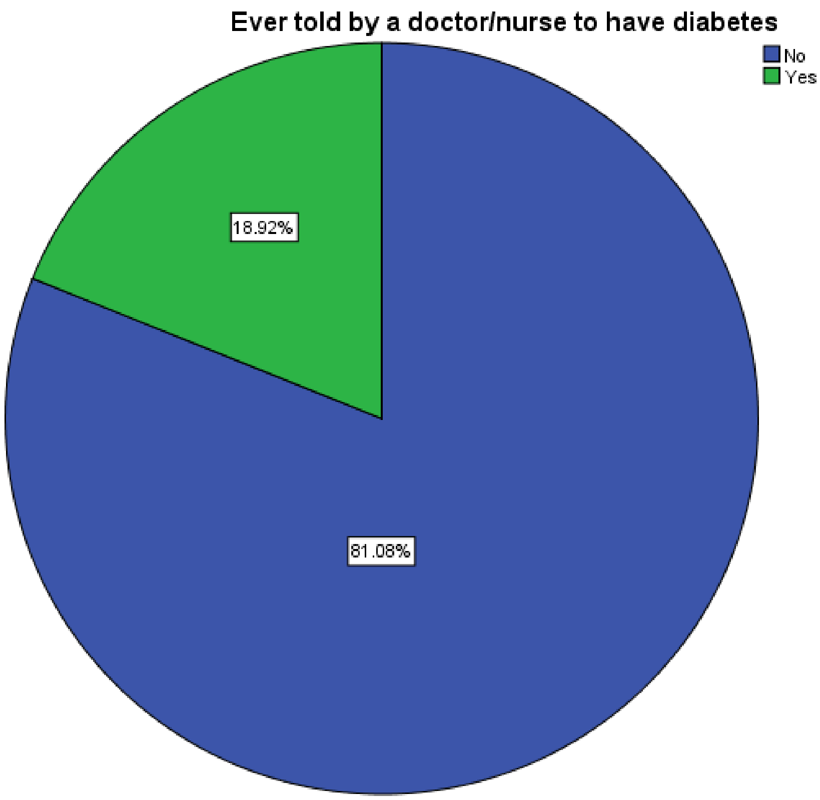
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Never married	583	5.8	5.8	5.8
	Married	8400	83.3	83.3	89.1
	Widowed	943	9.3	9.3	98.4
	Divorced	160	1.6	1.6	100.0
	Total	10086	100.0	100.0	



In terms of current marital status, out of 10,086 participants, 583 were never married, 8,400 were married, 943 were widowed, and 160 were divorced. In terms of percentage, it can be observed that the majority of participants were married at 83.3%, followed by widowed at 9.3%, never married at 5.8%, and the lowest category by divorce at 1.6%.

Ever told by a doctor/nurse to have diabetes

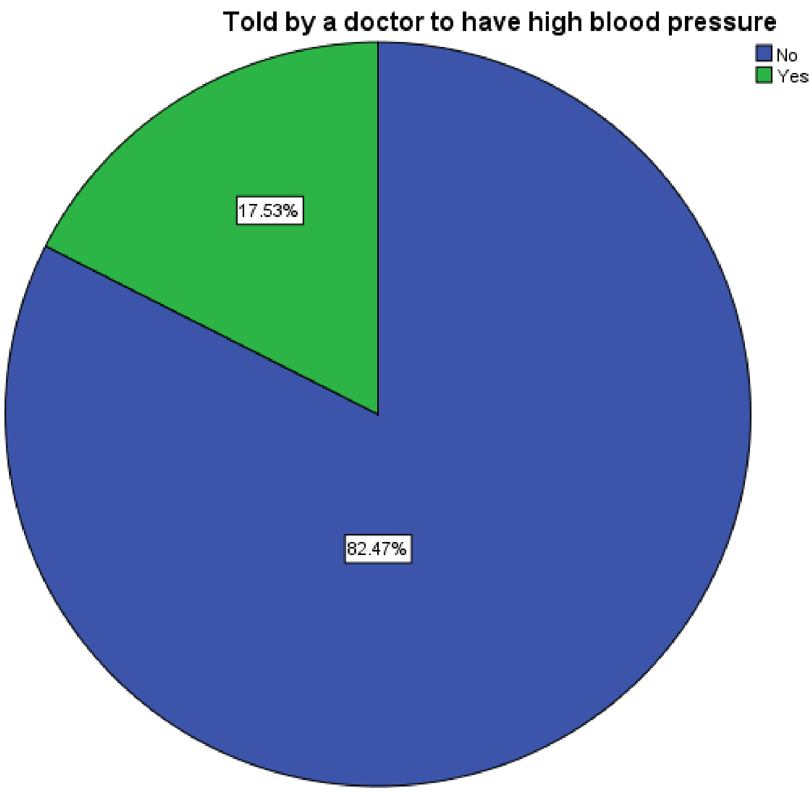
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	2558	25.4	81.1	81.1
	Yes	597	5.9	18.9	100.0
	Total	3155	31.3	100.0	
Missing	9	1	.0		
	System	6930	68.7		
	Total	6931	68.7		
Total		10086	100.0		



In terms of being told by a healthcare professional they have diabetes, out of 3,155 valid responses, 2,558 had not been told and 597 had. In terms of percentage, it can be observed that the majority of participants had not been told they have diabetes at 81.1% (valid percent), while the "Yes" category accounted for 18.9%.

Told by a doctor to have high blood pressure

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	8318	82.5	82.5	82.5
	Yes	1768	17.5	17.5	100.0
	Total	10086	100.0	100.0	

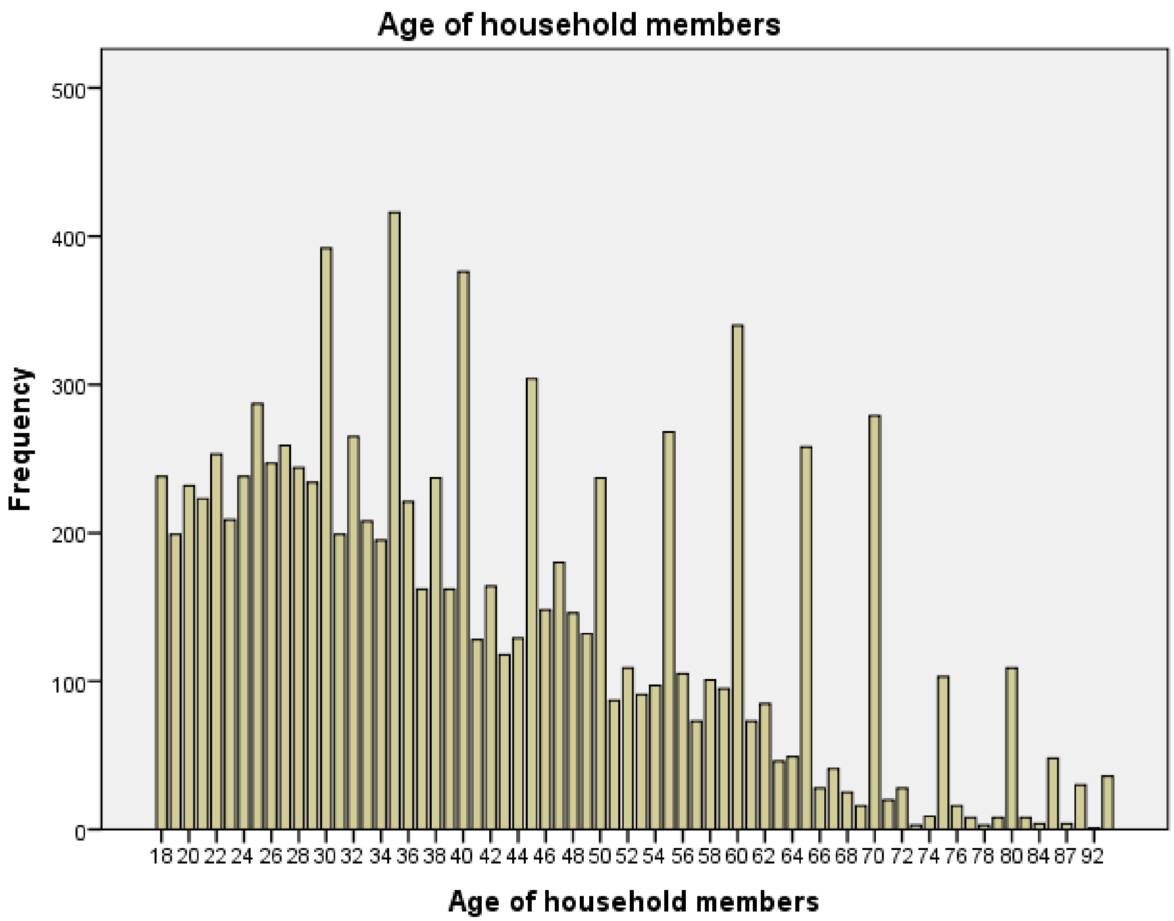


In terms of being told by a doctor they have high blood pressure, out of 10,086 participants, 8,318 had not been told and 1,768 had. In terms of percentage, it can be observed that the majority of participants had not been told they have high blood pressure at 82.5%, while the "Yes" category accounted for 17.5%.

Age of household members

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18	238	2.4	2.4	2.4
	19	199	2.0	2.0	4.3
	20	232	2.3	2.3	6.6
	21	223	2.2	2.2	8.8
	22	253	2.5	2.5	11.4
	23	209	2.1	2.1	13.4
	24	238	2.4	2.4	15.8
	25	287	2.8	2.8	18.6
	26	247	2.4	2.4	21.1
	27	259	2.6	2.6	23.6
	28	244	2.4	2.4	26.1
	29	234	2.3	2.3	28.4
	30	392	3.9	3.9	32.3
	31	199	2.0	2.0	34.2
	32	265	2.6	2.6	36.9
	33	208	2.1	2.1	38.9
	34	195	1.9	1.9	40.9
	35	416	4.1	4.1	45.0
	36	221	2.2	2.2	47.2
	37	162	1.6	1.6	48.8
	38	237	2.3	2.3	51.1
	39	162	1.6	1.6	52.7
	40	376	3.7	3.7	56.5
	41	128	1.3	1.3	57.7
	42	164	1.6	1.6	59.4
	43	118	1.2	1.2	60.5
	44	129	1.3	1.3	61.8
	45	304	3.0	3.0	64.8
	46	148	1.5	1.5	66.3
	47	180	1.8	1.8	68.1
	48	146	1.4	1.4	69.5
	49	132	1.3	1.3	70.8
	50	237	2.3	2.3	73.2
	51	87	.9	.9	74.1
	52	109	1.1	1.1	75.1
	53	91	.9	.9	76.0
	54	97	1.0	1.0	77.0
	55	268	2.7	2.7	79.7
	56	105	1.0	1.0	80.7
	57	73	.7	.7	81.4
	58	101	1.0	1.0	82.4
	59	95	.9	.9	83.4
	60	340	3.4	3.4	86.7
	61	73	.7	.7	87.5
	62	85	.8	.8	88.3
	63	46	.5	.5	88.8
	64	49	.5	.5	89.2
	65	258	2.6	2.6	91.8
	66	28	.3	.3	92.1
	67	41	.4	.4	92.5
	68	25	.2	.2	92.7
	69	16	.2	.2	92.9
	70	279	2.8	2.8	95.7

71	20	.2	.2	95.9
72	28	.3	.3	96.1
73	3	.0	.0	96.2
74	9	.1	.1	96.3
75	103	1.0	1.0	97.3
76	16	.2	.2	97.4
77	8	.1	.1	97.5
78	3	.0	.0	97.5
79	8	.1	.1	97.6
80	109	1.1	1.1	98.7
82	8	.1	.1	98.8
84	4	.0	.0	98.8
85	48	.5	.5	99.3
87	4	.0	.0	99.3
90	30	.3	.3	99.6
92	1	.0	.0	99.6
95+	36	.4	.4	100.0
Total	10086	100.0	100.0	



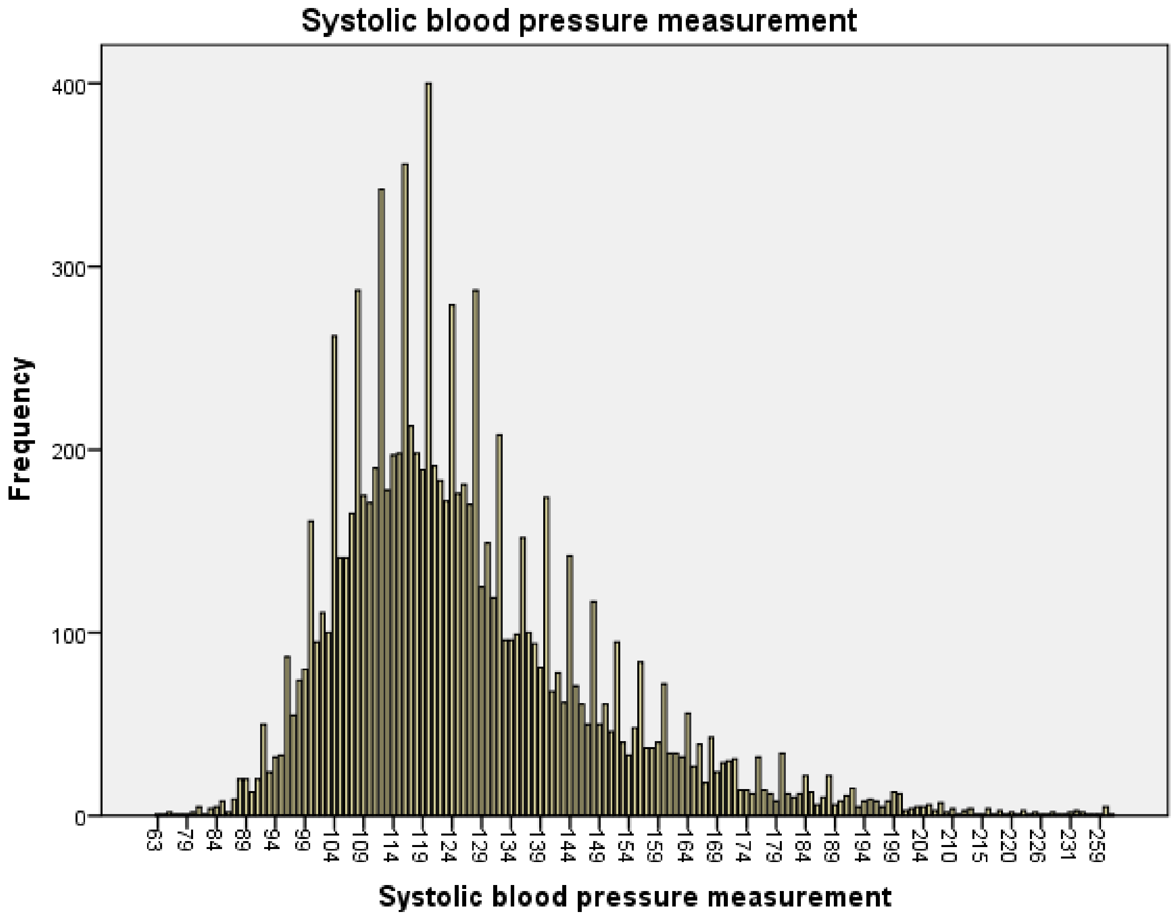
In terms of the age of household members, out of 10,086 valid responses, 5,154 participants were aged 38 or younger and 4,932 participants were aged 39 or older. In terms of percentage, it can be observed that the majority of participants were 38 years old or younger at 51.1% (cumulative percent), while the category of 39 years and older accounted for 48.9%.

Systolic blood pressure measurement

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	63	1	.0	.0	.0
	70	1	.0	.0	.0
	72	2	.0	.0	.0
	73	1	.0	.0	.0
	78	1	.0	.0	.1
	79	1	.0	.0	.1
	80	2	.0	.0	.1
	81	5	.0	.0	.1
	82	1	.0	.0	.1
	83	4	.0	.0	.2
	84	5	.0	.0	.2
	85	8	.1	.1	.3
	86	2	.0	.0	.3
	87	9	.1	.1	.4
	88	20	.2	.2	.6
	89	20	.2	.2	.8
	90	13	.1	.1	1.0
	91	20	.2	.2	1.2
	92	50	.5	.5	1.6
	93	24	.2	.2	1.9
	94	32	.3	.3	2.2
	95	33	.3	.3	2.5
	96	87	.9	.9	3.4
	97	55	.5	.5	3.9
	98	74	.7	.7	4.7
	99	80	.8	.8	5.5
	100	161	1.6	1.6	7.1
	101	95	.9	.9	8.0
	102	111	1.1	1.1	9.1
	103	100	1.0	1.0	10.1
	104	262	2.6	2.6	12.7
	105	141	1.4	1.4	14.1
	106	141	1.4	1.4	15.5
	107	165	1.6	1.6	17.1
	108	287	2.8	2.8	20.0
	109	175	1.7	1.7	21.7
	110	171	1.7	1.7	23.4
	111	190	1.9	1.9	25.3
	112	342	3.4	3.4	28.7
	113	178	1.8	1.8	30.4
	114	197	2.0	2.0	32.4
	115	198	2.0	2.0	34.4
	116	356	3.5	3.5	37.9
	117	213	2.1	2.1	40.0
	118	198	2.0	2.0	42.0
	119	189	1.9	1.9	43.8
	120	400	4.0	4.0	47.8
	121	191	1.9	1.9	49.7
	122	183	1.8	1.8	51.5
	123	172	1.7	1.7	53.2
	124	279	2.8	2.8	56.0

125	176	1.7	1.7	57.7
126	181	1.8	1.8	59.5
127	170	1.7	1.7	61.2
128	287	2.8	2.8	64.0
129	125	1.2	1.2	65.3
130	149	1.5	1.5	66.8
131	119	1.2	1.2	67.9
132	208	2.1	2.1	70.0
133	96	1.0	1.0	71.0
134	96	1.0	1.0	71.9
135	99	1.0	1.0	72.9
136	152	1.5	1.5	74.4
137	100	1.0	1.0	75.4
138	94	.9	.9	76.3
139	81	.8	.8	77.1
140	174	1.7	1.7	78.9
141	68	.7	.7	79.5
142	78	.8	.8	80.3
143	62	.6	.6	80.9
144	142	1.4	1.4	82.3
145	71	.7	.7	83.0
146	61	.6	.6	83.6
147	50	.5	.5	84.1
148	117	1.2	1.2	85.3
149	50	.5	.5	85.8
150	61	.6	.6	86.4
151	46	.5	.5	86.8
152	95	.9	.9	87.8
153	40	.4	.4	88.2
154	33	.3	.3	88.5
155	48	.5	.5	89.0
156	84	.8	.8	89.8
157	37	.4	.4	90.2
158	37	.4	.4	90.6
159	40	.4	.4	90.9
160	72	.7	.7	91.7
161	34	.3	.3	92.0
162	34	.3	.3	92.3
163	32	.3	.3	92.7
164	56	.6	.6	93.2
165	27	.3	.3	93.5
166	39	.4	.4	93.9
167	18	.2	.2	94.0
168	43	.4	.4	94.5
169	24	.2	.2	94.7
170	29	.3	.3	95.0
171	30	.3	.3	95.3
172	31	.3	.3	95.6
173	14	.1	.1	95.7
174	14	.1	.1	95.9
175	12	.1	.1	96.0
176	32	.3	.3	96.3
177	14	.1	.1	96.5
178	12	.1	.1	96.6
179	8	.1	.1	96.6
180	34	.3	.3	97.0
181	12	.1	.1	97.1

182	10	.1	.1	97.2
183	12	.1	.1	97.3
184	22	.2	.2	97.5
185	13	.1	.1	97.7
186	6	.1	.1	97.7
187	10	.1	.1	97.8
188	22	.2	.2	98.0
189	6	.1	.1	98.1
190	8	.1	.1	98.2
191	11	.1	.1	98.3
192	15	.1	.1	98.4
193	5	.0	.0	98.5
194	8	.1	.1	98.6
195	9	.1	.1	98.7
196	8	.1	.1	98.7
197	5	.0	.0	98.8
198	8	.1	.1	98.9
199	13	.1	.1	99.0
200	12	.1	.1	99.1
201	3	.0	.0	99.1
202	4	.0	.0	99.2
203	5	.0	.0	99.2
204	5	.0	.0	99.3
205	6	.1	.1	99.3
206	3	.0	.0	99.4
208	7	.1	.1	99.4
209	2	.0	.0	99.5
210	4	.0	.0	99.5
211	1	.0	.0	99.5
212	3	.0	.0	99.5
213	4	.0	.0	99.6
214	1	.0	.0	99.6
215	1	.0	.0	99.6
216	4	.0	.0	99.6
217	1	.0	.0	99.7
218	3	.0	.0	99.7
219	1	.0	.0	99.7
220	2	.0	.0	99.7
221	1	.0	.0	99.7
222	3	.0	.0	99.8
224	1	.0	.0	99.8
225	2	.0	.0	99.8
226	1	.0	.0	99.8
227	1	.0	.0	99.8
228	2	.0	.0	99.8
229	1	.0	.0	99.8
230	1	.0	.0	99.8
231	2	.0	.0	99.9
232	3	.0	.0	99.9
233	2	.0	.0	99.9
234	1	.0	.0	99.9
243	1	.0	.0	99.9
259	1	.0	.0	99.9
Refused	5	.0	.0	100.0
Other	1	.0	.0	100.0
Total	10086	100.0	100.0	



Finally, in terms of systolic blood pressure measurement, out of 10,080 valid responses, 5,191 participants had a reading of 122 mmHg or lower and 4,889 participants had a reading of 123 mmHg or higher. In terms of percentage, it can be observed that the majority of participants had a systolic blood pressure measurement of 122 mmHg or lower at 51.5% (cumulative percent), while the category of 123 mmHg and higher accounted for 48.5%.

5.2. Cross Tabulations with bar chart & Chi-square test Analysis

5.2.1. Crosstab with Division ✖ High Blood Pressure

Crosstab

			Told by a doctor to have high blood pressure		
			No	Yes	Total
Division	Barisal	Count	861	229	1090
		% within Division	79.0%	21.0%	100.0%
		% within Told by a doctor to have high blood pressure	10.4%	13.0%	10.8%
		% of Total	8.5%	2.3%	10.8%
	Chittagong	Count	1085	253	1338
		% within Division	81.1%	18.9%	100.0%
		% within Told by a doctor to have high blood pressure	13.0%	14.3%	13.3%
		% of Total	10.8%	2.5%	13.3%
	Dhaka	Count	1095	226	1321
		% within Division	82.9%	17.1%	100.0%
		% within Told by a doctor to have high blood pressure	13.2%	12.8%	13.1%
		% of Total	10.9%	2.2%	13.1%
	Khulna	Count	1204	267	1471
		% within Division	81.8%	18.2%	100.0%
		% within Told by a doctor to have high blood pressure	14.5%	15.1%	14.6%
		% of Total	11.9%	2.6%	14.6%
	Mymensingh	Count	987	172	1159
		% within Division	85.2%	14.8%	100.0%
		% within Told by a doctor to have high blood pressure	11.9%	9.7%	11.5%
		% of Total	9.8%	1.7%	11.5%
	Rajshahi	Count	1130	208	1338
		% within Division	84.5%	15.5%	100.0%
		% within Told by a doctor to have high blood pressure	13.6%	11.8%	13.3%
		% of Total	11.2%	2.1%	13.3%
	Rangpur	Count	1037	191	1228
		% within Division	84.4%	15.6%	100.0%
		% within Told by a doctor to have high blood pressure	12.5%	10.8%	12.2%
		% of Total	10.3%	1.9%	12.2%
	Sylhet	Count	919	222	1141
		% within Division	80.5%	19.5%	100.0%
		% within Told by a doctor to have high blood pressure	11.0%	12.6%	11.3%
		% of Total	9.1%	2.2%	11.3%
Total		Count	8318	1768	10086

% within Division	82.5%	17.5%		100.0%
% within Told by a doctor to have high blood pressure	100.0%	100.0%		100.0%
% of Total	82.5%	17.5%		100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	27.133[a]	7	.000
Likelihood Ratio	27.058	7	.000
Linear-by-Linear Association	6.347	1	.012
N of Valid Cases	10086		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 191.07.

State of Hypotheses:

Null Hypothesis (H_0): There is no association between division and high blood pressure.

Alternative Hypothesis (H_A): There is an association between division and high blood pressure

Level of Significance:

The level of significance was set at $\alpha = 0.05$ or 5%.

Test Statistics:

From the SPSS output:

Pearson Chi-Square Value = 27.133

Degrees of Freedom (df) = 7

Asymptotic Significance (p-value) = 0.000

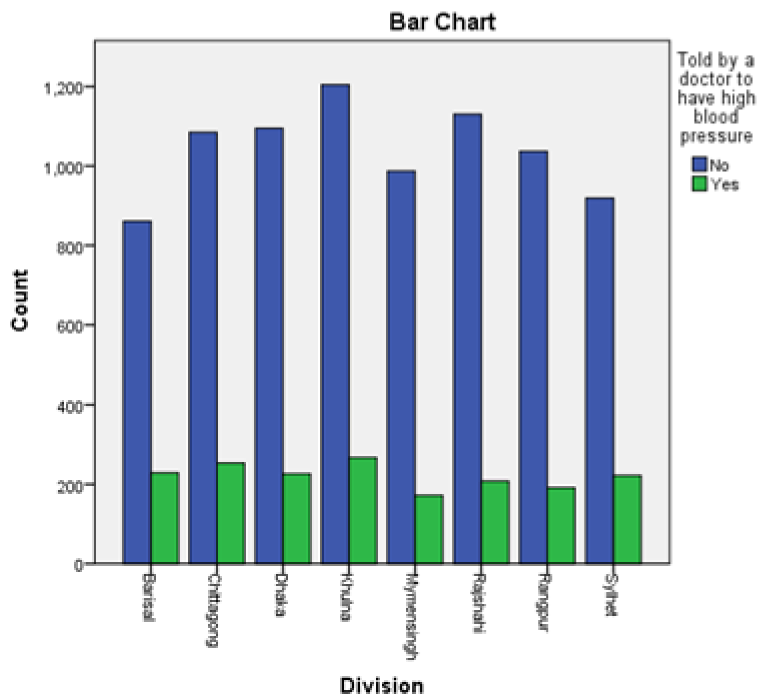
Decision Rule:

If p-value < α (0.05), we reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05

Conclusion:

The null hypothesis is rejected. There is significant association between division and high blood pressure.



The bar chart shows the distribution of respondents by division based on whether they were told by a doctor that they have high blood pressure. In all divisions, the number of individuals not diagnosed with high blood pressure is much higher than those who were diagnosed. Across all divisions, most respondents do not have high blood pressure since about 82.5% reported no and 17.5% reported affirmative out of 10,086 people. Regional differences are evident since Barisal and Sylhet show the highest proportions of high blood pressure, whereas Mymensingh has the lowest. Although Khulna has the highest number of cases in absolute terms, this reflects larger sample size instead of higher proportion.The Chi-square test indicates that these differences are statistically significant (p = 0.000), confirming an association between division of residence and high blood pressure status.

5.2.2. Crosstab with Types of Residence ✖ High Blood Pressure

Crosstab

			Told by a doctor to have high blood pressure	
			No	Yes
Type of place of residence	Urban	Count	3064	734
		% within Type of place of residence	80.7%	19.3%
		% within Told by a doctor to have high blood pressure	36.8%	41.5%
		% of Total	30.4%	7.3%
	Rural	Count	5254	1034
		% within Type of place of residence	83.6%	16.4%
		% within Told by a doctor to have high blood pressure	63.2%	58.5%
		% of Total	52.1%	10.3%
Total		Count	8318	1768
		% within Type of place of residence	82.5%	17.5%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%
		% of Total	82.5%	17.5%

Crosstab

			Total
Type of place of residence	Urban	Count	3798
		% within Type of place of residence	100.0%
		% within Told by a doctor to have high blood pressure	37.7%
		% of Total	37.7%
	Rural	Count	6288
		% within Type of place of residence	100.0%
		% within Told by a doctor to have high blood pressure	62.3%
		% of Total	62.3%
Total		Count	10086
		% within Type of place of residence	100.0%
		% within Told by a doctor to have high blood pressure	100.0%
		% of Total	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	13.604[a]	1	.000		
Continuity Correction[b]	13.405	1	.000		
Likelihood Ratio	13.473	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	13.602	1	.000		
N of Valid Cases	10086				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 665.76.

b. Computed only for a 2x2 table

State of Hypotheses:

Null Hypothesis (H_0): There is no association between types of residence and high blood pressure.

Alternative Hypothesis (H_A): There is an association between types of residence and high blood pressure.

Level of Significance:

The level of significance was set at $\alpha = 0.05$ or 5%.

Test Statistics:

From the SPSS output:

Pearson Chi-Square Value = 13.604

Degrees of Freedom (df) = 1

Asymptotic Significance (p-value) = 0.000

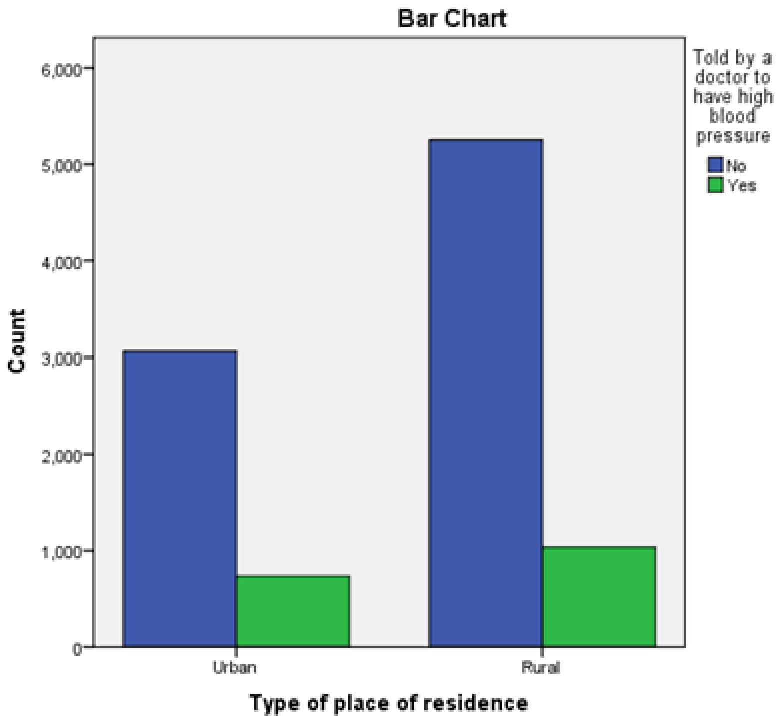
Decision Rule:

If p-value < α (0.05), we reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion:

The null hypothesis is rejected. There is significant association between types of residence and high blood pressure.



In the study conducted, the bar chart shows differences in high blood pressure between urban and rural residents. Although rural areas have more total cases due to a larger population size, urban residents have a higher prevalence of high blood pressure which is 19.3% compared to rural residents having 16.4%. Among 1768 individuals with high blood pressure, 58.5% are from rural areas and 41.5% are from urban areas. This reflects the larger rural sample rather than a higher risk. The Chi-square test ($\chi^2 = 13.604$, $p = 0.000$) confirms that the residence type is significantly associated with high blood pressure.

5.2.3. Crosstab with Current Marital Status ✖ High Blood Pressure

Crosstab

			Told by a doctor to have high blood pressure	
			No	Yes
Current marital status	Never married	Count	564	19
		% within Current marital status	96.7%	3.3%
		% within Told by a doctor to have high blood pressure	6.8%	1.1%
		% of Total	5.6%	0.2%
	Married	Count	7013	1387
		% within Current marital status	83.5%	16.5%
		% within Told by a doctor to have high blood pressure	84.3%	78.5%
		% of Total	69.5%	13.8%
	Widowed	Count	603	340
		% within Current marital status	63.9%	36.1%
		% within Told by a doctor to have high blood pressure	7.2%	19.2%
		% of Total	6.0%	3.4%
	Divorced	Count	138	22
		% within Current marital status	86.3%	13.8%
		% within Told by a doctor to have high blood pressure	1.7%	1.2%
		% of Total	1.4%	0.2%
Total		Count	8318	1768
		% within Current marital status	82.5%	17.5%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%
		% of Total	82.5%	17.5%

Crosstab

			Total
Current marital status	Never married	Count	583
		% within Current marital status	100.0%
		% within Told by a doctor to have high blood pressure	5.8%
		% of Total	5.8%
	Married	Count	8400
		% within Current marital status	100.0%
		% within Told by a doctor to have high blood pressure	83.3%
		% of Total	83.3%

	Widowed	Count	943
		% within Current marital status	100.0%
		% within Told by a doctor to have high blood pressure	9.3%
		% of Total	9.3%
	Divorced	Count	160
		% within Current marital status	100.0%
		% within Told by a doctor to have high blood pressure	1.6%
		% of Total	1.6%
Total		Count	10086
		% within Current marital status	100.0%
		% within Told by a doctor to have high blood pressure	100.0%
		% of Total	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	313.594[a]	3	.000
Likelihood Ratio	307.447	3	.000
Linear-by-Linear Association	216.811	1	.000
N of Valid Cases	10086		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 28.05.

State of Hypotheses:

Null Hypothesis (H_0): There is no association between current marital status and high blood pressure.

Alternative Hypothesis (H_A): There is an association between current marital status and high blood pressure.

Level of Significance:

The level of significance was set at $\alpha = 0.05$ or 5%.

Test Statistics:

From the SPSS output:

Pearson Chi-Square Value = 393.594

Degrees of Freedom (df) = 3

Asymptotic Significance (p-value) = 0.000

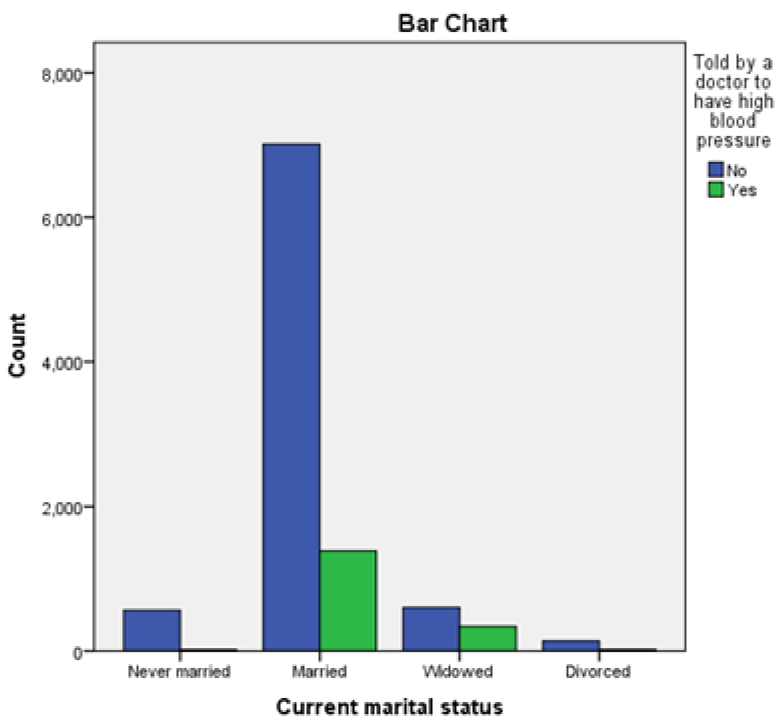
Decision Rule:

If p-value < α (0.05), we reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion:

The null hypothesis is rejected. There is a significant association between types of residence and high blood pressure.



In the study conducted, high blood pressure prevalence varies markedly by marital status. Widowed individuals have the highest prevalence with 36.1%, followed by married with 16.5% and divorced with 13.8%, while never-married individuals show the lowest prevalence with 3.3%. Although the widowed group has the highest rate, married individuals dominate the sample with 8,400 out of 10,086 individuals, resulting in the tallest bars and accounting for

78.5% of all high blood pressure cases, whereas widowed individuals contribute 19.2% of cases. The association between marital status and high blood pressure is highly statistically significant ($\chi^2 = 313.594$, $p = 0.000$), indicating a strong link between marital status and the likelihood of having high blood pressure.

5.2.4. Crosstab with Diabetes Status ✖ High Blood Press ure

Crosstab

			Told by a doctor to have high blood pressure		
			No	Yes	Total
Ever told by a doctor/nurse to have diabetes	No	Count	1924	634	2558
		% within Ever told by a doctor/nurse to have diabetes	75.2%	24.8%	100.0%
		% within Told by a doctor to have high blood pressure	85.4%	70.4%	81.1%
		% of Total	61.0%	20.1%	81.1%
	Yes	Count	330	267	597
		% within Ever told by a doctor/nurse to have diabetes	55.3%	44.7%	100.0%
		% within Told by a doctor to have high blood pressure	14.6%	29.6%	18.9%
		% of Total	10.5%	8.5%	18.9%
Total		Count	2254	901	3155
		% within Ever told by a doctor/nurse to have diabetes	71.4%	28.6%	100.0%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%	100.0%
		% of Total	71.4%	28.6%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	94.316[a]	1	.000		
Continuity Correction[b]	93.341	1	.000		
Likelihood Ratio	88.580	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	94.286	1	.000		
N of Valid Cases	3155				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 170.49.

b. Computed only for a 2x2 table

State of Hypotheses:

Null Hypothesis (H_0): There is no association between diabetes status and high blood pressure.

Alternative Hypothesis (H_A): There is an association between diabetes status and high blood pressure.

Level of Significance:

The level of significance was set at $\alpha = 0.05$ or 5%.

Test Statistics:

From the SPSS output:

Pearson Chi-Square Value = 94.316

Degrees of Freedom (df) = 1

Asymptotic Significance (p-value) = 0.000

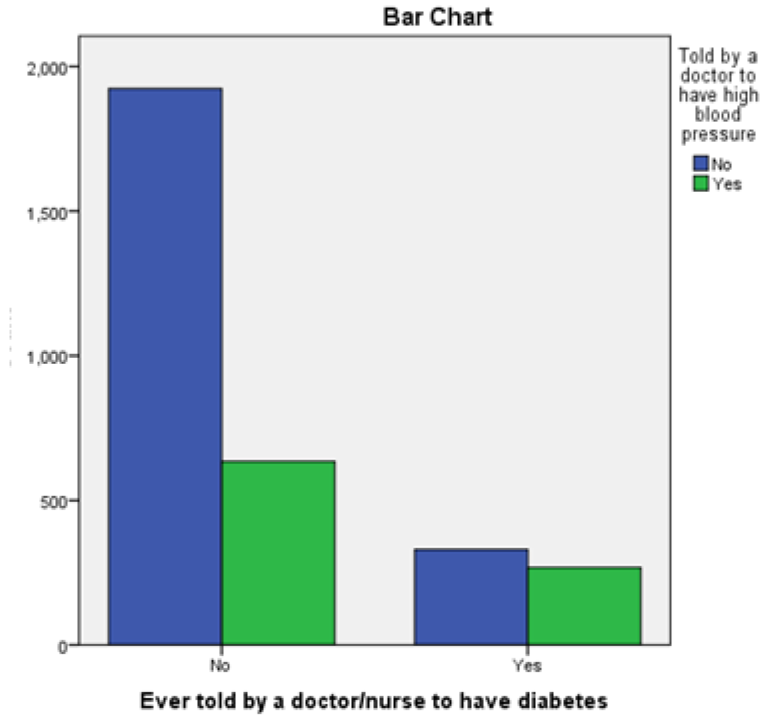
Decision Rule:

If p-value < α (0.05), we reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion:

The null hypothesis is rejected. There is significant association between diabetes status and high blood pressure.



In the study, the results show a strong association between diabetes and high blood pressure. Among individuals with diabetes 44.7% also have high blood pressure, compared to 24.8% among those without diabetes indicating that people with diabetes are nearly twice as likely to have high blood pressure. Out of the 901 individuals with high blood pressure, 70.4% are non-diabetic and 29.6% are diabetic. However, diabetics are over-represented given they comprise only 18.9% of the total sample. The relationship is highly statistically significant (χ^2 = 94.316, p = 0.000), confirming a strong link between diabetes status and high blood pressure.

5.2.5. Crosstab with Education ✖ High Blood Pressure

Cross Tab

			Told by a doctor to have high blood pressure		Total
			No	Yes	
Highest educational level attained	No education, preschool	Count	1847	567	2414
		% within Highest educational level attained	76.5%	23.5%	100.0%
		% within Told by a doctor to have high blood pressure	22.2%	32.1%	23.9%
	Primary	Count	2448	541	2989
		% within Highest educational level attained	81.9%	18.1%	100.0%
		% within Told by a doctor to have high blood pressure	29.4%	30.6%	29.6%
	Secondary	Count	2553	432	2985
		% within Highest educational level attained	85.5%	14.5%	100.0%
		% within Told by a doctor to have high blood pressure	30.7%	24.4%	29.6%
	Higher	Count	1467	228	1695
		% within Highest educational level attained	86.5%	13.5%	100.0%
		% within Told by a doctor to have high blood pressure	17.6%	12.9%	16.8%
	Don't know	Count	3	0	3
		% within Highest educational level attained	100.0%	0.0%	100.0%
		% within Told by a doctor to have high blood pressure	0.0%	0.0%	0.0%
Total		Count	8318	1768	10086
		% within Highest educational level attained	82.5%	17.5%	100.0%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	99.393[a]	4	.000
Likelihood Ratio	97.745	4	.000
Linear-by-Linear Association	91.421	1	.000
N of Valid Cases	10086		

a. 2 cells (20.0%) have expected count less than 5. The minimum expected count is .53.

State the Hypotheses

<u>Null Hypothesis (H_0):</u> There is no association between education level and high blood pressure.

<u>Alternative Hypothesis (H_A):</u> There is an association between education level and high blood pressure.

Determine the Significance Level

The level of significance was set at $\alpha = 0.05$ or 5%

Test Statistics From the SPSS output:

Pearson Chi-Square Value = 99.39

Degrees of Freedom (df) = 4

Asymptotic Significance (p-value) = 0.000

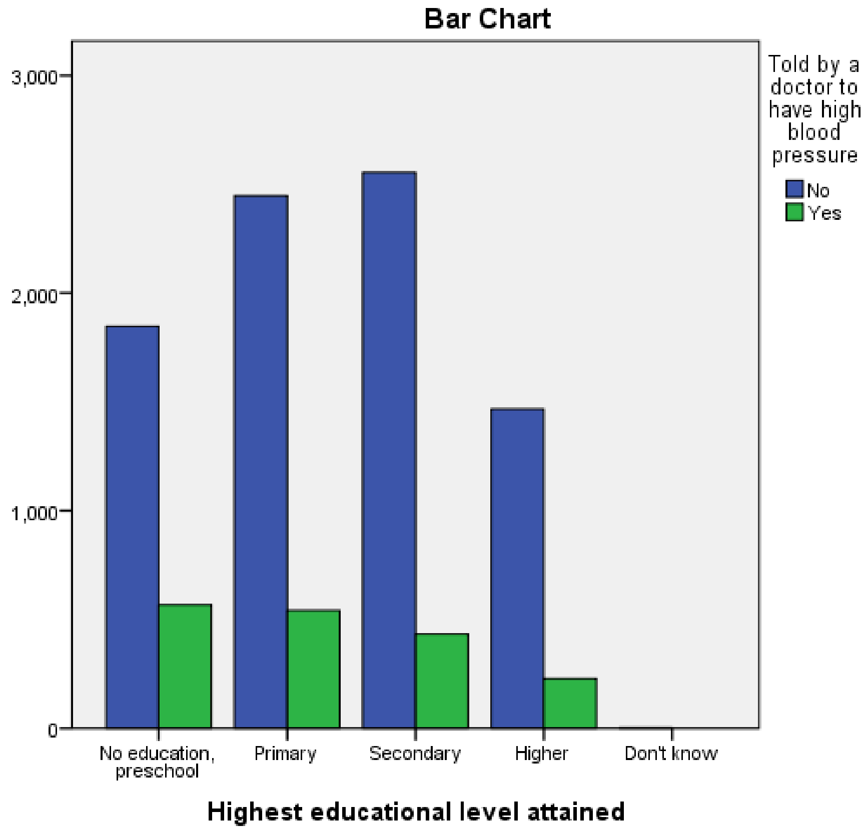
Decision Rule

If p-value < α (0.05), reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion

The null hypothesis is rejected. There is a statistically significant association between education level and high blood pressure.



The relationship between high blood pressure and education level was investigated in the study population. 23.5% of people with no education or only a preschool education reported having high blood pressure, compared to 76.5% who did not. A significant percentage of all individuals with high blood pressure came from this group.

18.1% of people with only a primary education had high blood pressure, compared to 81.9% of people without any education, showing a reduced prevalence. There was a further decline in the prevalence of high blood pressure in the secondary education group, with 14.5% having it and 85.5% not.

Higher educated participants had the lowest percentage of high blood pressure (13.5%), with 86.5% having no high blood pressure. Overall, the bar graph shows a distinct decreasing pattern in cases of high blood pressure as educational level rises. This implies that a higher risk of high blood pressure is linked to a lower educational level.

5.2.6. Crosstab with Wealth Index ✖ High Blood Pressure

Cross Tab

			Told by a doctor to have high blood pressure		Total
			No	Yes	
Wealth index combined	Poorest	Count	1529	219	1748
		% within Wealth index combined	87.5%	12.5%	100.0%
		% within Told by a doctor to have high blood pressure	18.4%	12.4%	17.3%
	Poorer	Count	1543	259	1802
		% within Wealth index combined	85.6%	14.4%	100.0%
		% within Told by a doctor to have high blood pressure	18.6%	14.6%	17.9%
	Middle	Count	1627	334	1961
		% within Wealth index combined	83.0%	17.0%	100.0%
		% within Told by a doctor to have high blood pressure	19.6%	18.9%	19.4%
	Richer	Count	1629	379	2008
		% within Wealth index combined	81.1%	18.9%	100.0%
		% within Told by a doctor to have high blood pressure	19.6%	21.4%	19.9%
	Richest	Count	1990	577	2567
		% within Wealth index combined	77.5%	22.5%	100.0%
		% within Told by a doctor to have high blood pressure	23.9%	32.6%	25.5%
Total		Count	8318	1768	10086
		% within Wealth index combined	82.5%	17.5%	100.0%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	88.983[a]	4	.000
Likelihood Ratio	89.455	4	.000
Linear-by-Linear Association	87.713	1	.000
N of Valid Cases	10086		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 306.41.

State the Hypotheses

Null Hypothesis (H_0): There is no association between wealth index and high blood pressure.

Alternative Hypothesis (H_A): There is an association between wealth index and high blood pressure.

Determine the Significance Level

The level of significance was set at α = 0.05 or 5%

Test Statistics

<u>From the SPSS output:</u>

Pearson Chi-Square Value = 88.98

Degrees of Freedom (df) = 4

Asymptotic Significance (p-value) = 0.000

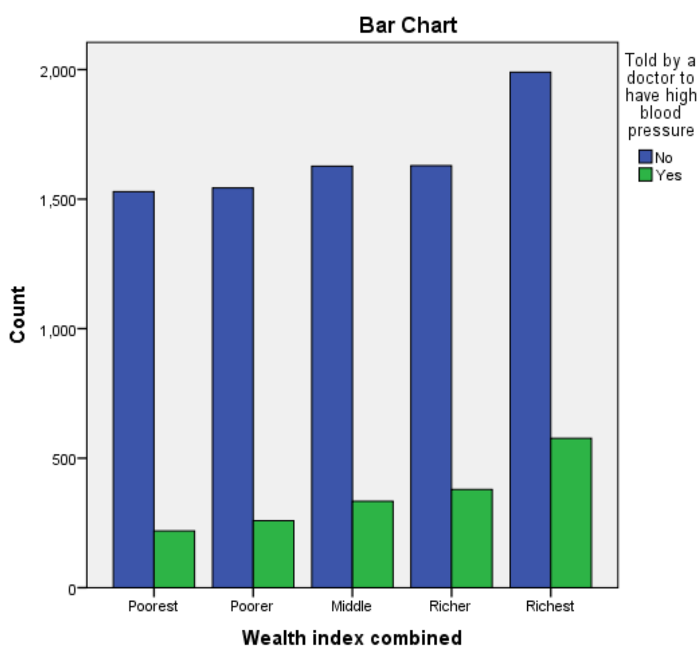
Decision Rule

If p-value < 0.05, reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion

The null hypothesis is rejected. There is a statistically significant association between wealth index and high blood pressure.



There are observable differences between wealth categories in the bar graph that illustrate the relationship between high blood pressure and the wealth index. 12.5% of those in the poorest category had high blood pressure, compared to 87.5% who did not. The percentage of people in the poorer category who reported having high blood pressure rose to 14.4%, whereas 85.6% did not.

17.0% of people in the middle to lower wealth category had high blood pressure, while 83.0% did not, showing a continuous increase. 81.1% of the wealthier group did not have high blood pressure, compared to 18.9%. With 22.5% reporting high blood pressure and 77.5% reporting no high blood pressure, the wealthiest group had the highest prevalence.

Overall, the bar chart shows a steady increase in the prevalence of high blood pressure as wealth status increases, indicating that those in wealthier groups report higher levels of high blood pressure.

5.2.7. Crosstab with Smoking ✖ High Blood Pressure

Cross Tab

			Told by a doctor to have high blood pressure		
			No	Yes	Total
Smoked	No	Count	7058	1406	8464
		% within Smoked	83.4%	16.6%	100.0%
		% within Told by a doctor to have high blood pressure	84.9%	79.5%	84.0%
	Yes	Count	1254	362	1616
		% within Smoked	77.6%	22.4%	100.0%
		% within Told by a doctor to have high blood pressure	15.1%	20.5%	16.0%
Total		Count	8312	1768	10080
		% within Smoked	82.5%	17.5%	100.0%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	31.446[a]	1	.000		
Continuity Correction[b]	31.047	1	.000		
Likelihood Ratio	29.840	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	31.443	1	.000		
N of Valid Cases	10080				

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 283.44.

b. Computed only for a 2x2 table

State the Hypotheses

Null Hypothesis (H_0): There is no association between smoking status and high blood pressure.

Alternative Hypothesis (H_A): There is an association between smoking status and high blood pressure.

Determine the Significance Level

The significance level was set at $\alpha = 0.05$ or 5%

Test Statistics

From the SPSS output:

Pearson Chi-Square Value = 31.45

Degrees of Freedom (df) = 1

Asymptotic Significance (p-value) = 0.000

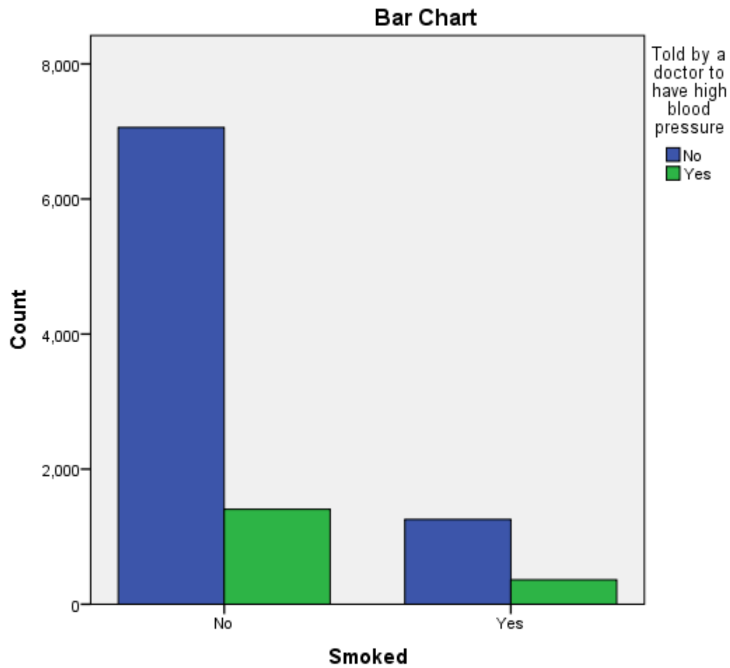
Decision Rule

If p-value < 0.05, reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion

The null hypothesis is rejected. There is a statistically significant association between smoking status and high blood pressure.



Using a bar chart, the relationship between high blood pressure and smoking status was investigated. Of those who did not smoke, 16.6% had high blood pressure, whereas 83.4% did not. The majority of participants belonged to this category.

On the other hand, 22.4% of smokers said they had high blood pressure, while 77.6% did not. Smoking may have a negative impact on blood pressure levels, as seen by the much higher percentage of high blood pressure among smokers compared to non-smokers.

It is seen in the bar chart that smokers had taller bars for high blood pressure, indicating that smoking is linked to an increased risk of high blood pressure.

5.2.8. Crosstab with Caffeinated Drinks ✖ High Blood Pressure

Cross Tab

			Told by a doctor to have high blood pressure		Total
			No	Yes	
Had caffeinated drink	No	Count	7718	1564	9282
		% within Had caffeinated drink	83.2%	16.8%	100.0%
		% within Told by a doctor to have high blood pressure	92.8%	88.5%	92.1%
	Yes	Count	595	204	799
		% within Had caffeinated drink	74.5%	25.5%	100.0%
		% within Told by a doctor to have high blood pressure	7.2%	11.5%	7.9%
Total		Count	8313	1768	10081
		% within Had caffeinated drink	82.5%	17.5%	100.0%
		% within Told by a doctor to have high blood pressure	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	38.344[a]	1	.000		
Continuity Correction[b]	37.746	1	.000		
Likelihood Ratio	34.959	1	.000		
Fisher's Exact Test				.000	.000
Linear-by-Linear Association	38.341	1	.000		
N of Valid Cases	10081				

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 140.13.

b. Computed only for a 2x2 table

State the Hypotheses

Null Hypothesis (H_0): There is no association between caffeinated drink consumption and high blood pressure.

Alternative Hypothesis (H_A): There is an association between caffeinated drink consumption and high blood pressure.

Determine the Significance Level

The significance level was set at $\alpha = 0.05$ or 5%

Test Statistics

<u>From the SPSS output:</u>

Pearson Chi-Square Value = 38.34

Degrees of Freedom (df) = 1

Asymptotic Significance (p-value) = 0.000

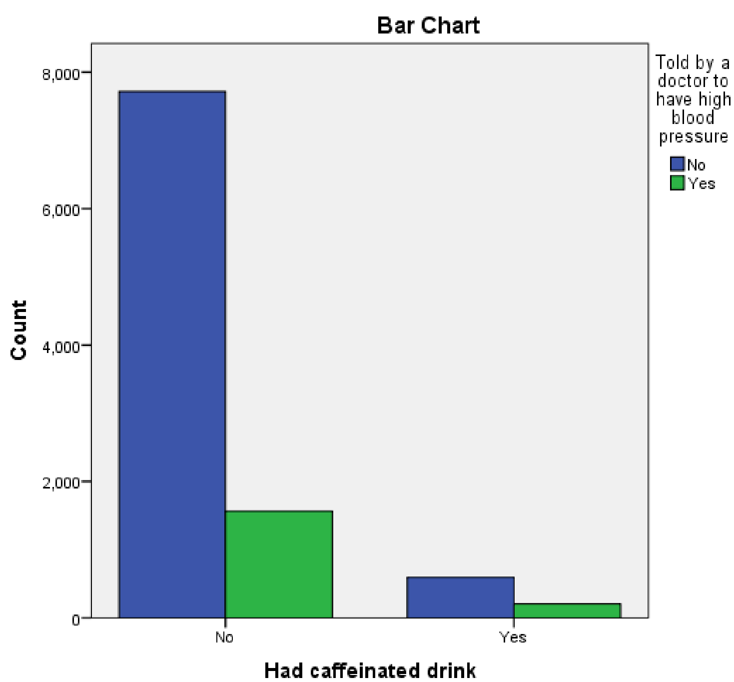
Decision Rule

If p-value < 0.05, reject the null hypothesis.

Here, p-value = 0.000, which is far less than 0.05.

Conclusion

The null hypothesis is rejected. There is a statistically significant association between caffeinated drink consumption and high blood pressure.



Differences between those who drank caffeinated beverages and those who did not are seen in the bar graph indicating the relationship between high blood pressure and caffeinated drink use. 16.8% of participants who avoided caffeinated beverages reported having elevated blood pressure, compared to 83.2% who did not.

Compared to 74.5% of people who did not drink caffeinated beverages, 25.5% of those who did reported having high blood pressure. This suggests that people who drank caffeinated beverages had a higher percentage of increased blood pressure.

Overall, the bar graph indicates that drinking caffeinated beverages is linked to an increased risk of high blood pressure, even though there isn't much of a difference between the two groups.

6. Dispersion and Distribution Analysis

The following table represents mean and the five number summaries for the following variables: Age of household members and Systolic blood pressure measurement

Statistics

		Age of household members	Systolic blood pressure measurement
N	Valid	10086	10086
	Missing	0	0
Mean		41.14	126.99
Minimum		18	63
Maximum		95	996
Percentiles	25	28.00	111.00
	50	38.00	122.00
	75	52.00	137.00

This descriptive statistics output summarizes the target variables from the nationally representative Bangladesh Demographic and Health Survey (BDHS) 2022, providing mean and five number summaries of two variables: Age of household members and Systolic blood pressure measurement.

The variable age of household members (N = 10086) does not have any missing values and spans a wide range, reflecting a diverse age group. The mean age for 10,086 participants is 41.14, indicating that on average, household members are about 41.14 years old. The minimum age of the household member is 18 years and the maximum age is 95 years.

For the variable age of household members, the first quartile (Q1) is 28 which shows that 25% of the household member’s age is less than or equal to 28 years old. The second quartile (Q2) is 38 indicating that 50% of the household members are 38 years or younger and the third quartile (Q3) is 52, where 75% of the members are 52 years or younger. This age distribution is fairly spread out with most individuals being middle-aged adults and the mean is 41.14 which is slightly higher than the median (Q2), suggesting that it is a right skewed distribution.

The average of the variable systolic blood pressure (BP) measurement of 10,086 individuals is 126.99, which is approximately 127 mmHg. The minimum systolic BP measurement is 63 mmHg, indicating very low readings, possibly from hypotensive individuals. The maximum reading reported is 996 mmHg, which is not realistic data and represents a data entry or measurement error.

For the variable systolic blood pressure measurement, the 25th percentile (Q1) is 111, where 25% of the individuals have systolic BP less than or equal to 111 mmHg. The 50th percentile (Q2) is 122, where 50% of the individuals have a reading less than or equal to 122 mmHg and the 75th percentile (Q3) is 137, where 75% of the individuals have systolic BP less than or equal to 137 mmHg.

Most participants have systolic BP within the normal range (normal systolic BP = 120 mmHg), but a substantial proportion falls into the pre-hypertensive range. Here, the mean is 127 which is greater than the median (Q2), indicating the distribution is right-tailed. The extreme maximum suggests the dataset should have the outliers removed before further analysis.

The following table summarizes the measures of variability and the shape of the distribution for both variables:

Descriptive Statistics

	N	Range	Mean	Std. Deviation	Variance	Skewness	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error
Age of household members	10086	77	41.14	16.298	265.622	.690	.024
Systolic blood pressure measurement	10086	933	126.99	30.827	950.278	13.694	.024
Valid N (listwise)	10086						

Interpretation of Results:

1. Age of household members:

- **Dispersion:** The ages of the 10,086 participants vary by a range of 77 years. The Standard Deviation of 16.298 indicates that, on average, individual ages deviate from the mean (41.14) by about 16 years, suggesting a relatively wide spread of ages across the household members.
- **Shape (Skewness):** The skewness statistic is 0.690.
  - Since this value is positive, the distribution is right-skewed (positively skewed).
  - Because the value is between 0.5 and 1.0, the distribution is considered moderately skewed. This means there is a slightly longer tail on the right side of the distribution (toward older ages).

2. Systolic blood pressure (SBP) measurement:

- **Dispersion:** The SBP shows an extreme Range of 933 and a high Standard Deviation of 30.827. This indicates very high variability in blood pressure readings within this sample.
- **Shape (Skewness):** The skewness statistic is 13.694.
  - This indicates an extreme positive skew.
  - A value this high suggests that the majority of the data points are clustered at the lower end (near the mean of 126.99), but there are extreme outliers on the high end that are pulling the distribution tail significantly to the right.

Variable	Skewness Statistic	Distribution Shape	Interpretation
Age	0.690	Moderately Positive Skew	The tail extends toward older ages; relatively balanced.
Systolic BP	13.694	Highly Positive Skew	Extreme outliers present on the high end of BP readings.

While the age distribution is relatively standard for a household survey, the systolic blood pressure data is highly skewed and variable, likely due to the presence of extreme clinical cases or data outliers within the BDHS 2022 dataset.

7. **Correlation & Regression:**

Correlations

		Systolic blood pressure measurement	Age of household members
Systolic blood pressure measurement	Pearson Correlation	1	.308**
	Sig. (2-tailed)		.000
	N	10086	10086
Age of household members	Pearson Correlation	.308**	1
	Sig. (2-tailed)	.000	
	N	10086	10086

**. Correlation is significant at the 0.01 level (2-tailed).

Correlation Coefficient:
Systolic blood pressure-Age = 0.308

The Pearson correlation coefficient ranges from -1 to +1:

- r=0, indicates no relation.
- r= +1, indicated perfect positive correlation
- r= -1, indicates perfect negative correlation.

Strength:

- 0.00- 0.19 = very weak
- 0.20-0.29 = weak
- 0.30-0.39 = weak to moderate
- 0.40-0.59 = moderate
- 0.60-0.79 = strong

Thus, Systolic blood pressure-Age = 0.308 **(weak to moderate positive correlation)**

Interpretation:
The correlation analysis has revealed a weak to moderate positive relationship between age and systolic blood pressure ($r= 0.308$, $p < 0.01$). This interprets that with increasing age the systolic blood pressure tends to increase slightly. However, the strength of the correlation is weak to moderate, implying age alone does not affect or strongly determine the systolic blood pressure and there are other factors that also may affect blood pressure levels.

Regression Analysis:

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Age of household members[b]	.	Enter

a. Dependent Variable: Systolic blood pressure measurement

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.308[a]	.095	.094	29.334

a. Predictors: (Constant), Age of household members

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	906220.051	1	906220.051	1053.126	.000[b]
	Residual	8677330.958	10084	860.505		
	Total	9583551.009	10085			

a. Dependent Variable: Systolic blood pressure measurement

b. Predictors: (Constant), Age of household members

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	103.059	.793		129.935	.000	101.505	104.614
	Age of household members	.582	.018	.308	32.452	.000	.546	.617

a. Dependent Variable: Systolic blood pressure measurement

A simple linear regression analysis was conducted to analyze the effect of age on systolic blood pressure. The model showed a weak to moderate positive relationship between age and systolic blood pressure with correlation coefficient of 0.308.

The coefficient for determination (R^2= 0.095) indicated that age accounted for approximately 9.5% of the total variation in systolic blood pressure and the remaining variation can be explained by other variables. The adjusted R^2(0,094) suggested that the model was stable after adjustment.

The regression model is highly significant as indicated by the ANOVA results ($F=1053.126$, $p < 0.001$), confirming that age is a significant predictor of systolic blood pressure.

Examination of the regression coefficient for age (B= 0.582) reveals that for every one year increase in age, systolic blood pressure increased by approximately 0.582 mm-Hg and this effect was statistically significant ($p < 0.001$).

Overall, these results demonstrate that age has a meaningful positive influence on systolic blood pressure. Although it only explains a smaller portion of the total variation, indicating that other factors also contribute to blood pressure level.

Significance:

All the observed relationships were statistically significant at the $p<0.001$(2 tailed), providing strong evidence that the effects are not zero, However, given the large sample size (N= 10,086), part of this statistical significance is influenced by the number of observations, as even small effects can appear significant in large datasets.

Therefore, it is important to consider the effect size when interpreting the results. Age shows a meaningful relationship with systolic blood pressure, although it accounts for only a portion of the variation, indicating the other factors also contribute

8. Conclusion

The overall data of the frequency tables show that the analysis of 10,086 participants across eight divisions reveals a predominantly rural (62.3%) , married (83.3%) , and younger demographic (51.1% are 38 or younger). Education is led by primary and secondary levels (29.6% each) , while the "Richest" wealth category is the most prominent at 25.5%. Health data indicates high blood pressure (17.5%) and diabetes (18.9% of respondents) prevalence, alongside low rates of smoking (16%) and caffeine use (7.9%).

The mean systolic blood pressure measured was seen to be highly positively skewed, which means the data contained extreme outliers that may distort analysis. So, there is a potential need for data transformation.

The analysis shows that high blood pressure is not evenly distributed across the population. Its prevalence varies significantly by geographic division, place of residence, marital status, and diabetes status.

The bar charts indicate that lower education level, higher wealth status, smoking, and consumption of caffeinated drinks are associated with higher proportions of high blood pressure. These graphical findings visually support the results of the Chi-square tests.

9. Reference

https://dhsprogram.com/